Navya Kumar

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EDUCATION

Columbia University (New York, USA) ***June 2025 – August 2025***
- Real Analysis: A (97%).

University of Cambridge (Cambridge, UK) ***October 2021 – June 2024***
- Bachelor of Economics degree: achieved high 2:1 overall (69%).
- Relevant Modules: Advanced Microeconomics (74%), Public Economics (72%), Advanced Macroeconomics (68%) and Industrial Organisation (66%)

King's College London Mathematics School (London, UK) ***September 2019 – June 2021***
- A Levels: Maths A*, Further Maths A*, Physics A*
- Research: Causes of the UK Productivity Puzzle, The Economic Impact of Global Food Waste

EMPLOYMENT EXPERIENCE

Columbia University (New York, USA) ***August 2024 – Present***
Pre-Doctoral Researcher
- Working for Professor Robert Metcalfe on multiple projects, ranging from environmental economics to real estate.

Bank of England (London, UK): Ranked in the top 10% of interns both summers.
Summer Intern ***June 2023 – August 2023***
- Completed research project in R, evaluated two datasets with 50 million daily entries about margin paid on financial derivatives for consistency. Liaised with staff to correct inconsistencies in reporting by clearing houses.
- Presented findings to over 30 people across the Bank, providing feedback as to how to monitor firms better.

Summer Intern ***June 2022 – August 2022***
- Conducted data project using word analysis in Python, evaluating survey responses about monthly price shifts of 500 SME firms. Assessed changes in causes of inflation; determined reliability of responses in forecasting inflation.
- Developed a programme with a reproducible code to track future survey answers to help with inflation now-casts; presented results to Monetary Analysis Team; findings were reviewed in pre-MPC meetings to set national rate.

WORKING PAPERS

The Impact of the Rise in Energy Prices during the UK Energy Crisis. [PDF]

LEADERSHIP

Cambridge Society for Economic Pluralism (London, UK)
President ***June 2023 – June 2024***
- Planned speaker events, debates and forums about economic issues with distinguished academics including Professor Daron Acemoglu, Ann Pettifor and Professor Branko Milanovic alongside PhD researchers.
- Recruited and managed a committee of 15 people, responsible for a membership base of 1600 students.
- Directed an essay competition for 60 prospective students, ultimately marked by Professor Charles Brendon.

Speakers Officer ***September 2022 – June 2023***
- Interviewed speakers including Professor Ha Joon Chang, Professor Richard Layard and Professor Gareth Austin.

Marshall Society (Cambridge, UK) ***July 2022 – June 2023***
Marshall Conference President
- Orchestrated a conference attended by 400 prospective economics students. Handled a budget of £5000. Featured talks by Cambridge professors including Professor Jose Gabriel Palma and Professor Jagjit Chadha.

Homerton College, University of Cambridge (Cambridge, UK) ***March 2022 – April 2022***
Telephone Campaign Officer
- Corresponded with alumni of Homerton College and raised £9687 in donations, compared with an average of £320.

RESEARCH ASSISTANT PROJECTS

Daryl Fairweather, Robert Metcalfe, Thi Mai Anh Nguyen
Allocation or Price: The Case of US Real Estate Agents

- Constructed national MLS transaction panel and a novel dataset covering buyer-side journeys.
- Evaluated incentives: how commission structures, partnerships, teams and promotions affect behavior, how refund structures affect buyers, manager and agent fixed effects.
- Contributed to the literature review and modeling the auction IO process of a buyer's journey.

Louise Bernard, Andy Hackett, Robert Metcalfe, Andrew Schein
Decarbonizing Heat: The Impact of Heat Pumps and a Time-of-Use Heat Pump Tariff on Energy Demand

- Calculated the marginal value of public funds of a £7500 heat pump subsidy in the UK and how this varies by temperature.

Louise Bernard, Andy Hackett, Robert Metcalfe, Luca Panzone, Andrew Schein
The Impact of Dynamic Prices on Electric Vehicle Public Charging Demand: Evidence from a Natural Field Experiment

- Estimated the consumer surplus of a reduction in the marginal price of electricity for EV charging, isolating the effects of substitution and demand creation.

Robert Metcalfe, Sefi Roth
Making the Invisible Visible: The Impact of Revealing Indoor Air Pollution on Behavior and Welfare

- Extracted and cleaned indoor air pollution data to estimate the impacts of revealing pollution measures to households in London. Quantified the overall welfare impact of providing a subsidy for indoor air pollution monitors and air purifiers.

Robert Metcalfe, Andrew Schein, Cohen Simpson, Yixin Sun
AI in Charge: Large-Scale Experimental Evidence on Electric Vehicle Charging Demand

- Evaluated the welfare gains of AI-managed EV charging, projecting potential grid savings from shifting electricity demand to non-peak hours.

Daniel Lopez Garcia, Robert Metcalfe, Andrew Schein, Yixin Sun
A Demand-Side Alternative to Renewable Curtailment: Natural Field Experimental Evidence from Two Countries

- Modeled the economic impact of avoiding curtailment and derived the bid markups demanded by generators with market power to curtail output.

Robert Hahn, Nathaniel Hendren, Robert Metcalfe, Ben Sprung-Keyser
A Welfare Analysis of Policies Impacting Climate Change

- Helped construct the welfare comparisons between regulations and subsidies: Corporate Average Fuel Economy standards (CAFE) vs a gas tax, Renewable Portfolio Standards (RPS) vs wind power production subsidies.

Luigi Butera, Andrew Kennedy, John List, Robert Metcalfe
Seeing Stars: Multiple Hypothesis Testing in Economics

- Read and classified papers into the type of MHT adjustments they required, which was used as training data for an AI model.

Xinming Du, Robert Metcalfe
Impact of Management on Maritime Productivity and Emissions

- Completed IRB and began initial analysis of the impact of software to reduce maritime emissions.

TECHNICAL EXPERTISE AND VOLUNTEERING

Technical Experience: Git, LaTeX, R, Python and Stata.
GRE: Quant: 170, Verbal: 163, Writing: 5.0.
Volunteering: Student Community Action Taskforce Volunteer 2022-24, Schools Connect Volunteer Tutor 2021-22, Cambridge Development Initiative Fundraising Support Officer 2021-22, YMCA Charity Shop Volunteer 2020-21
Languages: English (fluent), Hindi (native), Spanish (basic)